

Brains share the rule, but not the wiring

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Abstract

We trained a model on how electrical stimulation moves the brains of five mice, and then showed it a sixth mouse doing nothing special, just resting. From that resting activity, plus the settings of the stimulus we were about to deliver, the model had to say what would happen next in an animal it had never seen perturbed.

Two things came through. The model predicted what the mouse would do, meaning when it would notice the stimulus and report it, and how that changed with stimulation strength. It also predicted how the recorded population as a whole would respond, including the timing and the rise and fall. What it could not predict was the response of any individual neuron.

The reason turned out to be specific and measurable. The shared rule already gets each neuron's response shape right, meaning when the neuron fires, how its response ramps and decays, and how it grows with current. The only thing it gets wrong is how strongly each neuron responds, which is a single number per cell. Handing the model that one number per cell, and changing nothing else, raises the neural score from +0.035 to +0.422 in every one of the 6 mice. That number turned out to be essentially unguessable from resting activity, and we show why: low current stimulation does not light up a neat sphere of nearby cells, it grabs a sparse scattered set that depends on where that particular electrode happened to land. In a simulated cortex where we can switch that mechanism on and off, switching it on reproduces the failure, and recording more neurons at once repairs it.

So the shared dynamics that people describe across animals really is causal. It lives above the single cell scale. Individuality is in the neurons.

1 The question

Record the same brain area in two animals doing the same task and the population activity looks alike. Many methods now use that similarity, aligning recordings across sessions and subjects into a common space [1, 2, 5–7], and fitting latent dynamics that carry a behavioural decoder from one animal to another [3, 4].

All of that describes what activity looks like when the brain is left alone. It leaves open a question about mechanism. If two animals share a dynamical rule, then pushing them the same way should move them the same way. If the resemblance is only in the trajectories they happen to visit, then it need not.

We wanted a test that could tell those apart, so we set up a prediction problem. A model sees interventions in several animals. It then meets a new animal and is allowed to watch only its ordinary, unperturbed activity. Given the settings of a stimulus, it has to produce the whole time course of what follows, in the neurons and in the behaviour, before that stimulus is ever delivered to that animal.

2 What we measured

We used a public dataset of intracortical microstimulation in mouse somatosensory cortex [10]. 6 mice learned to report a stimulus delivered through a chronically implanted probe by turning a wheel. The stimulus is described by numbers we know in advance, which makes it easy to say what an unseen intervention means: the current, from 1 to 12 μA across 12 levels, and which contact on the probe delivered it, which sets its depth in cortex.

After quality control we kept 48 sessions, 949 sorted neurons, 16,532 stimulation trials and 13,044 windows of unperturbed activity, binned at 25 ms. Sessions record between 8 and 33 neurons at once, with a median of 19.

Three readouts run through the paper. The behavioural one is the probability that the animal has reported the stimulus by a given moment, which gives a smooth response curve that grows with current. The population one is the average firing rate across the recorded neurons. The finest one is the response of each neuron on its own.

We always score the effect of the stimulus rather than the raw activity, because a model can track activity well while knowing nothing about the intervention. Writing Δ for the measured change caused by the stimulus,

$$\Delta R^2 = 1 - \frac{\sum (\Delta - \hat{\Delta})^2}{\sum \Delta^2},$$

so a score of zero is exactly the model that says the stimulus does nothing, and one is perfect. Because Δ is estimated from a finite number of trials, we also compute how high any model could possibly score, by splitting trials in half and comparing the two halves. That ceiling sits at 0.91 for neurons and 0.84 for behaviour, so the measurements themselves are not what limits us.

Six animals is the sample size. The claim here is about animals. Several sessions from one mouse are still one mouse, so every headline number is inferred at the animal level. Confidence intervals come from a bootstrap that resamples animals and carries all of their sessions along, and significance comes from an exact sign flip permutation over the 6 animals. With six animals the smallest p value that test can ever return is 0.031, reached when all six fall on the same side, and we say so wherever it appears. Session level numbers appear alongside as secondary detail, and an earlier version of this work reported them as the headline, which overstated the evidence.

3 What transferred

3.1 Behaviour

The behavioural response transfers well. Taking the average response of the other five mice at each current and applying it to the sixth gives $\Delta R^2 = +0.572$ (95% CI [+0.451, +0.692]), positive in 6/6 animals, $p = 0.031$. A smooth curve fitted through current and time does about the same, at +0.591, and our dynamical model does about the same again, at +0.544. Figure 4 shows the measured and predicted curves for every held out mouse.

Two things are worth saying plainly about this. The effect is large and it holds in every animal, which is the strongest form the evidence can take with six of them. And the simplest method we could think of already achieves it. Averaging five mice predicts the sixth about as well as anything else we tried, and adding the new mouse's resting activity to the model made the prediction worse rather than better (+0.420).

The behavioural consequence of stimulating this part of cortex is conserved enough across individuals that it does not need a model of that individual.

3.2 Population activity

The population response carries the same shape from animal to animal. The correlation between the predicted and measured time course is $+0.710$ for the group average and $+0.669$ for the model, so the timing, the rise and fall, and the dependence on current all come across. The size of the effect is another matter. At the animal level the score is $+0.256 [-0.105, +0.580]$ for the group average and $+0.336 [+0.067, +0.579]$ for our best model, with 5/6 and 5/6 animals above zero and p values of 0.156 and 0.094. With six animals that is not enough to claim that population level effect sizes transfer reliably. The shape does, and the size is unresolved.

3.3 Our model does not beat the simple baselines

It is worth stating this plainly, because it is easy to hide. On every readout we compared the model against the best of the simple alternatives, using a paired test over animals. Nowhere does it win. For behaviour the difference is -0.047 against a smooth curve in current and time, at $p = 0.406$, and for the neural readouts the differences are similarly small and inconsistent in sign. The interesting quantity in this dataset turns out to be how conserved the response already is, rather than how well a dynamical model can exploit an individual's resting activity. Where the model does earn its place is in taking the shared rule apart, which is the next section.

3.4 Single neurons

Individual neurons do not transfer. The score is $+0.103$ for the average of the other mice (5/6 animals above zero, $p = 0.156$) and $+0.035$ for the dynamical model (3/6 animals, $p = 0.656$). This is not a limit of the measurement, since the ceiling for the same quantity is 0.91, and it is not a limit of one particular model, since a smooth function of physical variables, a manifold alignment style group average, and a model built from each animal's own spontaneous dynamics all land in the same place.

4 Where the failure comes from

Knowing that single neurons fail is less useful than knowing what is missing, so we took the shared rule apart.

One number per neuron. We froze the shared rule at the value fitted on the other animals and changed nothing about it. Then we allowed the held out animal one scalar per neuron, fitted on that animal's stimulation trials. This is not a prediction, because it uses data the protocol forbids. It is a way of asking what the shared rule already knows. A scalar per neuron cannot invent structure in time, and it cannot invent structure across currents, because each neuron's predicted time course and dose dependence are fixed by the shared rule.

That one number per neuron takes the score from $+0.035$ to $+0.422 [+0.355, +0.499]$, positive in 6/6 animals, with an animal level difference of $+0.387$ at $p = 0.031$, which is the floor for six animals.

Allowing an offset as well gives $+0.511$. So the shared rule was already getting the shape of each neuron’s response right. It was getting each neuron’s amplitude wrong.

That number is not guessable. Having found the missing quantity, we tried to predict it without touching any stimulation trial in the held out animal. The predictors were everything non interventional we had, including each neuron’s depth and its offset from each stimulating contact, its firing statistics, and its spontaneous coupling to the neurons near each contact, which is the natural linear response proxy for how strongly the stimulus should reach it. Shrinkage was chosen by a nested leave one animal out loop inside the training animals.

Across all 949 neurons, predicted and required amplitudes correlate at only $r = +0.174$, and applying the prediction moves the score from $+0.035$ to $+0.065$, which does not survive an animal level test ($p = 0.562$). An exact per neuron amplitude would give $+0.422$. So the gap is real, it is exactly one number per cell wide, and nothing we can observe without stimulating fills it.

Why it is not guessable. The obvious guess is distance. A neuron near the electrode should respond more. Pooled over every session and every condition, that is not what the data show. The correlation between how far a neuron sits from the stimulating contact and how much it responds is -0.013 , with no peak at the electrode and no falloff, just a broad elevation spread over more than a millimetre in both directions (Fig. 2b). This is the signature of low current stimulation driving axons of passage, which reaches a sparse and scattered set of cells rather than a dense local sphere, as direct imaging of stimulation evoked activity also reports [9]. Which cells those are is a property of where that particular electrode ended up in that particular brain. There is no species level rule to learn at that resolution.

5 A simulated cortex that tests the explanation

The explanation above is testable, so we built a cortex where we control the thing we think matters. Neurons sit along a depth axis and are connected by a rule shared across animals, plus a private part. Stimulation at a contact injects current, and we can choose who receives it.

In the *local* setting the electrode drives neurons by a smooth rule of distance, so which cells respond is the same rule in every animal. In the *sparse* setting the electrode drives a scattered subset drawn fresh for each animal and each contact, so the amount of drive is still shared while the identity of the recipients is private to that implant. We then vary how many neurons are recorded at once and score transfer exactly as we do for the mice.

The two settings come apart (Fig. 5). With a shared local rule the score averages $+0.679$ across population sizes, every animal is above zero at every size, and the score climbs steadily as more neurons are recorded, with a correlation of $+0.987$ between the score and the log of the population size. With private recruitment the average falls to $+0.409$, the trend with population size becomes erratic ($+0.582$), and the spread across animals roughly triples, from a mean interval width of 0.15 to 0.52. At the population sizes found in our mice the private setting is at its most unstable, with intervals that comfortably span zero.

Two things follow, and the second one is a correction to what we expected. Private recruitment does clearly damage single neuron transfer, costing about $+0.270$ in score at matched population size and making the result far less reliable from animal to animal. But it does not by itself abolish transfer in this simulation, so it is not a complete account of the failure in the mice, which is more severe. The honest reading is that private recruitment is one contributing cause among others, and that small simultaneously recorded populations are another.

The clean trend in the shared setting is the useful part for planning. When the rule governing who gets driven is shared, recording more neurons at once helps in a very regular way. Our mice have between 8 and 33 neurons per session, at the bottom of the range we simulated.

6 When can transfer work at all

It helps to know when this kind of transfer is possible in principle. Write the model as a shared latent state that every animal shares, a shared rule for how it evolves, a shared rule for how a stimulus pushes it, and an animal specific map from that state to the neurons we happen to record:

$$z_{t+1} = z_t + \Delta t [F(z_t, u_t) + G(z_t) a_t + R_i(z_t)], \quad (1)$$

$$y_t^{(i)} \sim \text{Poisson}(\text{softplus}(C_i z_t + b_i)), \quad \text{behaviour}_t = D(z_t). \quad (2)$$

Proposition 1 (Transfer is pinned down up to the symmetries of the shared flow). *Fix F and G . Suppose a new animal’s unperturbed law is matched both by (C_i, R_i) and by (C'_i, R'_i) , with C_i of full row rank and the latent state reachable under the exogenous inputs. Then there is an invertible T with $TF(z, u) = F(Tz, u)$ for all z and u , such that $C'_i = C_i T^{-1}$ and $R'_i = T R_i T^{-1}$. The two give predictions for the same intervention that differ by $C_i T^{-1} (TG(Tz)a - G(z)a)$. If the only such T is the identity, the predicted response is unique. Otherwise it is ambiguous by exactly that group, unless G happens to be equivariant under it.*

The argument is a change of variables $z \mapsto Tz$ in the two equations above, together with the requirement that the unperturbed law be preserved; full rank of C_i and reachability are what rule out reparameterisations that act only on directions the data never visit. Two things follow, and both showed up in practice. Resting activity sits near a fixed point, where the flow is weak and close to isotropic, so a freely fitted C_i has a lot of room and the new animal’s frame is poorly determined. That room can be closed by construction, by generating C_i from a shared function of per neuron features instead of fitting it freely, which is why a deliberately symmetric simulated teacher still transfers in our tests. The transfer is then only as good as those features are informative about how strongly the stimulus reaches a given cell, and in this preparation they are not.

7 How we kept ourselves honest

Several of the checks below changed a number in this paper, which is why they are worth listing.

The split half ceiling estimator was validated against simulations with known signal and noise across six regimes of noise level and trial count, agreeing with the attainable score to within 0.06.

Stimulation trials are delivered under a quiescence criterion, so inter trial windows sampled at random carry up to three times the pre stimulus firing rate of a real trial. Since predictions start from unperturbed initial conditions, that would have inflated every effect we report. Candidate windows are matched to stimulation trials on a joint quantile grid of pre stimulus firing rate and wheel speed and then trimmed to match on the mean, and any session still mismatched by more than a factor of 1.25 is dropped. The released audit reports 0 remaining problems.

Seeds are derived with a stable checksum rather than Python’s per process hash, so two independent builds of the analysis tensors agree exactly, and the results table reproduces bit for bit across re runs.

Deleting one middle current from the training animals turns out to be close to interpolating between its neighbours, so we also delete the top of the range, the bottom of the range, a contiguous block, whole

contact depths, and specific combinations of current and depth. Table 2 gives the behavioural result under each. Transfer to a genuinely unseen current is weaker than the interpolating case, and with six animals we do not claim it.

8 What this means

The similarity that people describe between animals is not only in the shape of the activity. Stimulating this part of cortex moves different mice in a way that is conserved enough to predict the behavioural consequence in an animal that has never been stimulated, and conserved enough to predict the shape of the population response. That is a statement about mechanism rather than appearance.

The same rule stops short of the individual cell, and it stops short in a specific way. It knows when a neuron will respond and how that response grows with current. It does not know how much any particular neuron will respond, and for electrical stimulation that is because the set of neurons the electrode reaches belongs to the implant rather than to the species.

For anyone calibrating a stimulation protocol, the practical reading is that the behavioural and population level effects seen in one animal are a reasonable guide to the next, and the effect on any named cell is not.

9 What we could not settle

Six animals is the binding constraint on almost everything here. The smallest p value available from an exact test over six animals is 0.031, so results that hold in all six are as strong as this design allows, and results that hold in four or five are suggestive at best. The population level effect sizes fall in the second category.

This is one species, one cortical area, one task and one way of perturbing. Electrical microstimulation is coarse and does not respect cell type, and an optogenetic perturbation that targets a defined population may well have more conserved structure at the single cell level than we found here.

The clearest next step follows from the simulation. Our account says single neuron transfer is limited by how well each animal’s own dynamics can be pinned down from resting activity, and that this improves with the number of neurons recorded at once. A dataset with hundreds of simultaneously recorded neurons and parameterised perturbations in many animals would settle it, and would also lift the animal count that limits the statistics here.

Data and code

The dataset is public [10]. Everything in this paper is rebuilt by `make all`, from downloading the raw files with a checksummed manifest through the leakage audit, the results table, the simulations, the figures and the numbers quoted in the text, which are generated directly from the result files.

References

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Table 1: What transfers to a held-out animal. Scores are averaged over animals, with a bootstrap over animals for the interval and an exact sign-flip permutation over animals for the p value. A score of zero is the model that says the stimulus does nothing. With six animals the smallest attainable p value is 0.031.

readout	method	ΔR^2	95% CI	r	animals > 0	p
behaviour	average of the other mice	+0.572	[+0.451, +0.692]	+0.652	6/6	0.031
	smooth curve in current and time	+0.591	[+0.477, +0.705]	+0.680	6/6	0.031
	shared rule (this paper)	+0.544	[+0.395, +0.674]	+0.660	6/6	0.031
	shared rule + resting activity	+0.420	[+0.204, +0.613]	+0.647	5/6	0.062
population	average of the other mice	+0.256	[-0.105, +0.580]	+0.710	5/6	0.156
	shared rule	+0.225	[-0.088, +0.494]	+0.669	4/6	0.188
	shared rule through own dynamics	+0.108	[-0.252, +0.446]	+0.668	3/6	0.562
	the same + predicted responsiveness	+0.336	[+0.067, +0.579]	+0.668	5/6	0.094
single neurons	average of the other mice	+0.103	[+0.010, +0.201]	+0.297	5/6	0.156
	shared rule	+0.093	[+0.010, +0.177]	+0.275	4/6	0.156
	shared rule through own dynamics	+0.035	[-0.078, +0.158]	+0.275	3/6	0.656
	the same + predicted responsiveness	+0.075	[-0.051, +0.192]	+0.275	5/6	0.281

Table 2: Behavioural transfer when the stimulus setting is also removed from every training animal. Deleting a middle current is close to interpolating between its neighbours; the other rows are harder.

held out from all training animals	ΔR^2	95% CI	animals > 0	p
nothing	+0.544	[+0.395, +0.674]	6/6	0.031
one middle current	-1.761	[-5.260, +0.283]	2/4	0.625
a block of three currents	+0.097	[-0.631, +0.658]	4/6	0.812
the top of the current range	+0.289	[-0.674, +0.788]	2/3	0.500
the bottom of the current range	-15.690	[-24.463, -6.756]	1/6	0.062
all superficial contacts	-55.259	[-68.460, -37.280]	0/5	0.062
all deep contacts	-14.756	[-24.174, -5.837]	1/5	0.125
high current at superficial contacts	-0.435	[-1.998, +0.635]	4/6	0.906

Figures

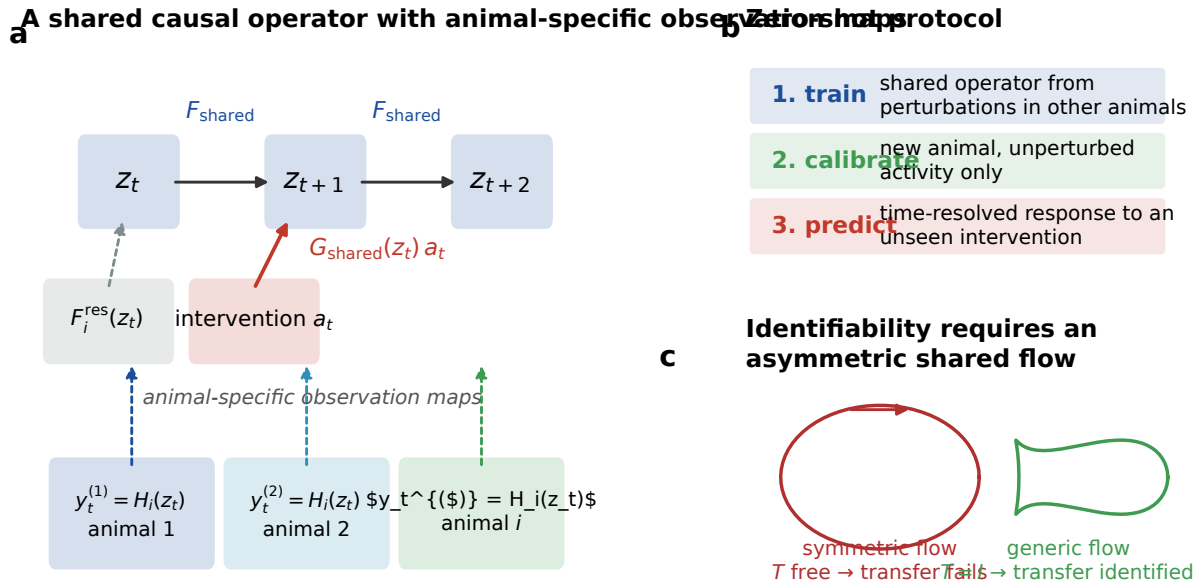


Figure 1: The question and the protocol. **a**, The model separates a rule shared by all animals, covering both how the latent state evolves and how a stimulus pushes it, from the animal specific map onto the neurons that happen to be recorded. **b**, The protocol. The shared rule is learned from stimulation in other animals, the new animal contributes only resting activity, and the response to a stimulus it has never received is then predicted. **c**, Proposition 1. A symmetric shared flow leaves a reparameterisation free and the prediction is ambiguous, while a generic flow pins the observation map down.

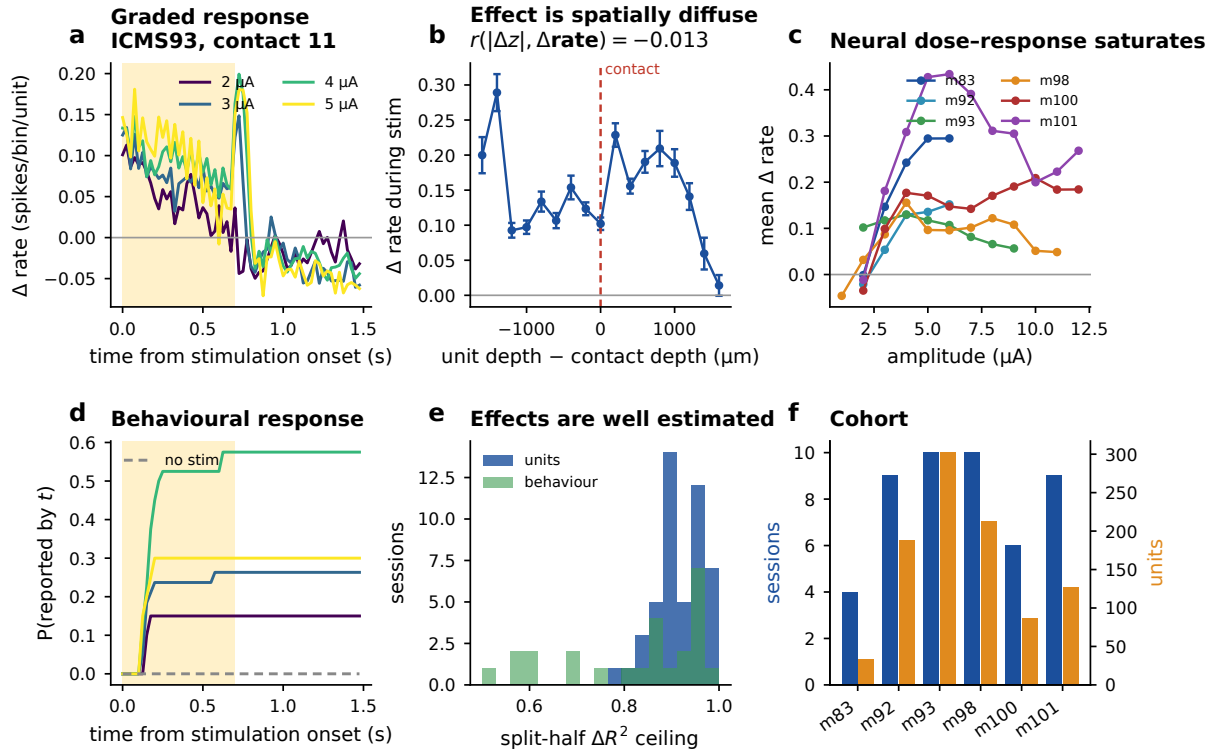


Figure 2: Microstimulation in mouse somatosensory cortex. **a**, The population response grows with current. **b**, Pooled over every session and condition, how much a neuron responds is essentially unrelated to how far it sits from the stimulating contact ($r = -0.013$). The response is spread out rather than centred on the electrode. **c**, Within animal dose response, which rises and then saturates. **d**, The behavioural response, meaning the probability of having reported the stimulus by time t . **e**, Split half ceilings for every session. **f**, Composition of the cohort.

Each dot is one mouse, held out in turn. Counts show how many mice scored above zero.

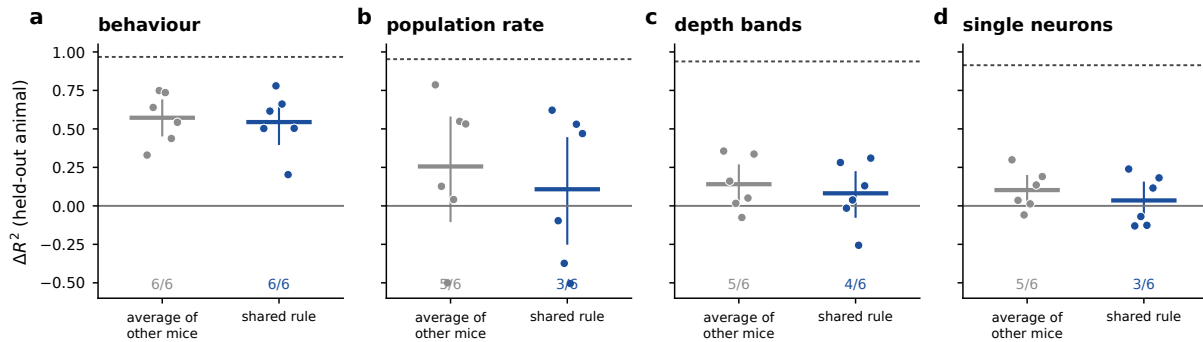


Figure 3: What transfers, by readout. Each point is one animal, held out in turn. Behaviour transfers in every animal. Population activity carries the right shape with unreliable size. Single neurons do not transfer. Dashed lines mark the highest score the trial counts allow.

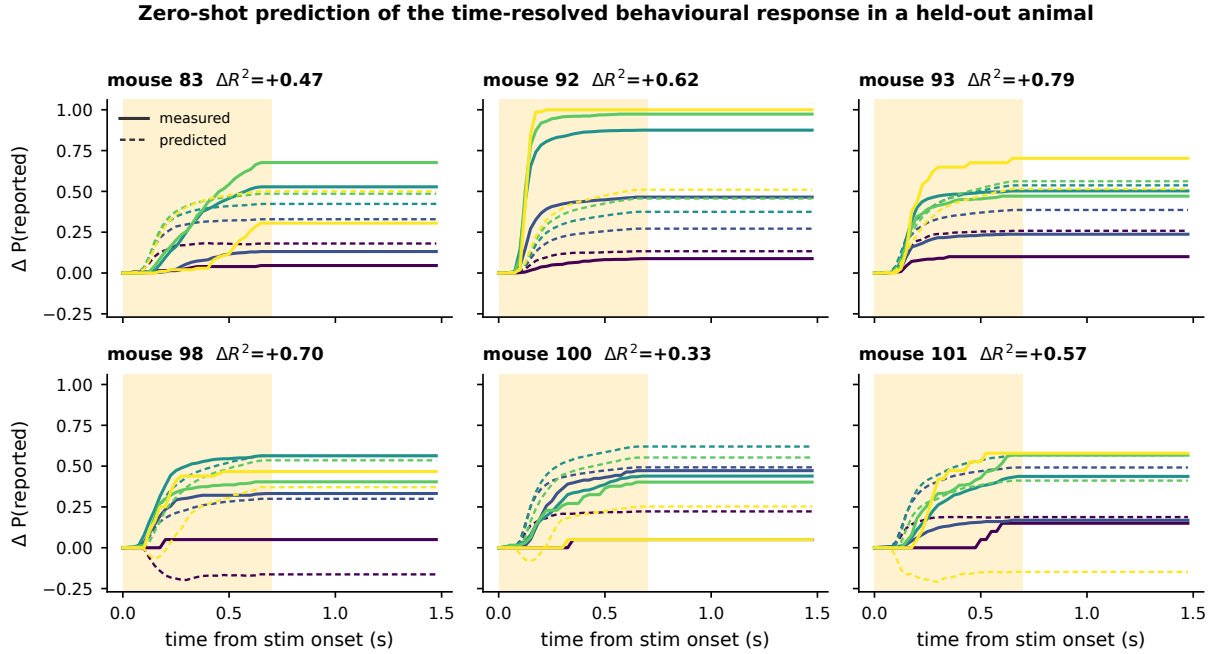


Figure 4: Predicting behaviour in a mouse the model never saw stimulated. Measured (solid) and predicted (dashed) change in the probability of having reported the stimulus, coloured by current. The model saw no stimulation trial from the animal shown in each panel.

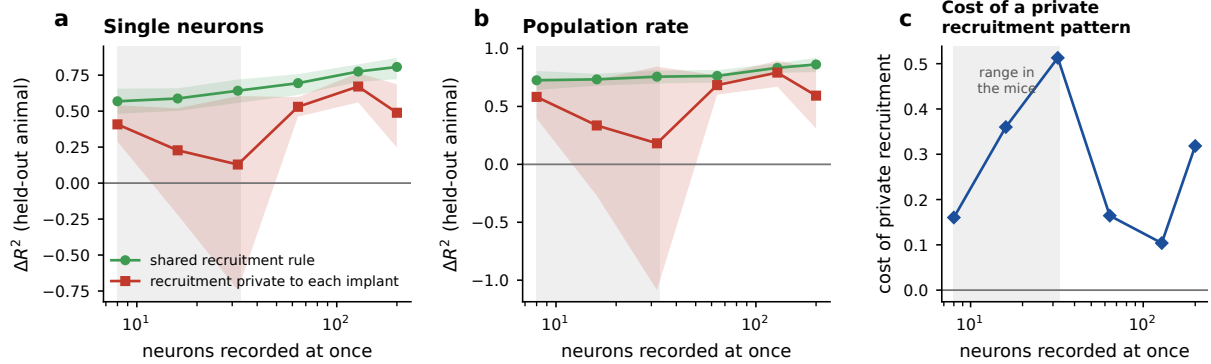


Figure 5: A simulated cortex tests the explanation. **a**, When the electrode drives neurons by a rule shared across animals, single neuron transfer works at every population size and improves as more neurons are recorded. When it drives a scattered set private to each implant, transfer fails at the population sizes found in the mice and recovers once enough neurons are recorded. **b**, The same comparison for the population readout, which is much less sensitive to how recruitment is arranged. **c**, The shaded band marks the range of simultaneously recorded neurons in the real dataset.